

AUSTO MOTOR COMPANY

Customer Purchase Behaviour Analysis Report

Submitted By

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1. Introduction

Austo Motor Company aims to understand customer purchasing behavior based on demographic, professional, and financial characteristics. Customer data analysis helps identify the factors that influence vehicle purchases and enables the company to make better marketing and business decisions.

Exploratory Data Analysis (EDA) is performed to identify trends, patterns, and relationships between customer attributes and vehicle types. The findings can help the company improve customer targeting, increase sales, and optimize promotional strategies.

2. Problem Statement

The company wants to answer the following business questions:

- Which type of customers purchase Hatchbacks, Sedans, and SUVs?
 - Does gender influence vehicle selection?
 - Which profession purchases more vehicles?
 - Does salary affect vehicle preference?
 - Do education and marital status influence purchasing behavior?
 - Does having personal or house loans impact buying decisions?
 - How does family income affect vehicle choice?
 - Which customer segment should the company target?
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3. Objectives

The objectives of this project are:

- Understand customer demographics.
 - Clean and prepare the dataset.
 - Perform Exploratory Data Analysis.
 - Identify relationships between variables.
 - Answer business questions.
 - Generate useful business recommendations.
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4. Dataset Description

The dataset contains customer information collected by Austo Motor Company.

Column	Description
Gender	Male/Female
Profession	Salaried/Business
Education	Graduate/Post Graduate
Marital Status	Married/Single
Personal Loan	Yes/No
House Loan	Yes/No
Partner Working	Yes/No
Salary	Customer Salary
Partner Salary	Partner Income
Total Salary	Family Income
No. of Dependents	Family Size
Make	Purchased Vehicle

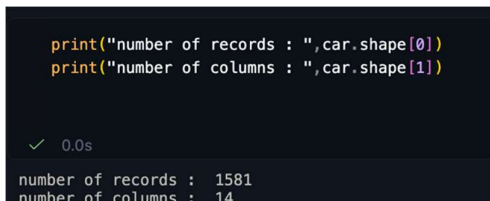
5. Data Profiling

Data profiling helps understand the dataset before analysis.

The following checks were performed:

- Dataset Shape
- Dataset Information
- Data Types
- Missing Values
- Duplicate Records
- Statistical Summary
- Unique Values

Dataset Shape



```
print("number of records : ",car.shape[0])
print("number of columns : ",car.shape[1])
```

✓ 0.0s

```
number of records : 1581
number of columns : 14
```

Interpretation

The dataset contains customer demographic and financial information used for the analysis.

Answer

The dataset contains sufficient records and variables required to analyze customer purchasing behavior.

The Dataset had 1581 records and 14 columns

Dataset Information

```

car.info()
✓ 0.0s

<class 'pandas.DataFrame'>
RangeIndex: 1581 entries, 0 to 1580
Data columns (total 14 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Age                   1581 non-null  int64
1   Gender                1528 non-null  str
2   Profession            1581 non-null  str
3   Marital_status       1581 non-null  str
4   Education             1581 non-null  str
5   No_of_Dependents     1581 non-null  int64
6   Personal_loan        1581 non-null  str
7   House_loan           1581 non-null  str
8   Partner_working      1581 non-null  str
9   Salary               1581 non-null  int64
10  Partner_salary       1475 non-null  float64
11  Total_salary         1581 non-null  int64
12  Price                1581 non-null  int64
13  Make                 1581 non-null  str
dtypes: float64(1), int64(5), str(8)
memory usage: 173.1 KB

```

Interpretation

The dataset contains both categorical and numerical variables required for EDA.

Answer

No major datatype issues were observed.

Missing Values

```

car.isnull().sum()

#check the missing value
✓ 0.0s

Age                0
Gender             53
Profession         0
Marital_status     0
Education          0
No_of_Dependents  0
Personal_loan      0
House_loan         0
Partner_working    0
Salary            0
Partner_salary     106
Total_salary       0
Price              0
Make               0
dtype: int64

```

Interpretation

The dataset was checked for null values.

Answer

We found missing value in two columns like Gender and Partner salary

→ Gender had 53 missing values

→ Partner salary had 106 missing value

Duplicate Values

```
car.duplicated().sum()

✓ 0.0s
np.int64(0)
```

Interpretation

Duplicate records were checked.

Answer

No duplicate records were found, ensuring reliable analysis.

Unique Values

```
#for finding unique of all coulmns at same time

car[['Gender', 'Profession', 'Marital_status', 'Education', 'Personal_loan', 'House_loan', 'Partner_working', 'Make']].apply(lambda x: x.unique())

✓ 0.0s
Gender          [Male, Femal, Female, nan, Femle]
Profession      [Business, Salaried]
Marital_status  [Married, Single]
Education       [Post Graduate, Graduate]
Personal_loan   [No, Yes]
House_loan      [No, Yes]
Partner_working [Yes, No]
Make            [SUV, Sedan, Hatchback]
dtype: object
```

Interpretation

Unique categories were verified for all categorical variables.

Answer

All categories are valid and suitable for visualization.

6. Data Cleaning

The following preprocessing steps were performed:

- Verified data types
- Checked missing values
- Checked duplicates
- Verified categorical values
- Prepared the dataset for visualization

The dataset was not clean it has some issues

→ Female spelling we wrong in Gender column in some rows so that we are replace the wrong spelling with right one

```
(variable) car: DataFrame
car.Gender = car.Gender.replace(["Femal","Femle"],["Female","Female"])

#replace incorect words (Femal,Femle) with correct word (Female)
✓ 0.0s
```

→ After that we had some null values in two columns

First in Gender :

→ we are solving it by most data had male so we are filling those null value with male

```
car.Gender = car.Gender.fillna("Male")

#replace missing value with male in Gender,because male is most frequent coustmer in this data
✓ 0.0s
```

Second in Partner salary;

→ fill the null values of partner salary by subtraction of salary with total salary

```
car.Partner_salary = car.Total_salary - car.Salary

# fill the null values of partner salary by subtraction of salary with total salary
✓ 0.0s
```

7. Exploratory Data Analysis (EDA)

Univariate Analysis

Gender Distribution



Objective

To understand the distribution of customers by gender.

Interpretation

The graph shows the number of male and female customers in the dataset.

Answer to Problem Statement

Both male and female customers contribute significantly to vehicle purchases. Therefore, marketing campaigns should target both genders while considering their individual preferences.

Profession Distribution



Objective

To identify the professional background of customers.

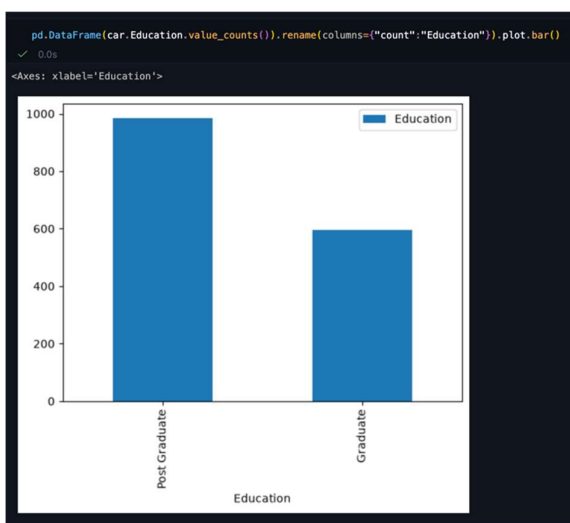
Interpretation

Customers mainly belong to Salaried and Business professions.

Answer

The majority of customers are salaried employees, indicating that they represent the company's primary customer base.

Education Distribution



Objective

To understand customers' education levels.

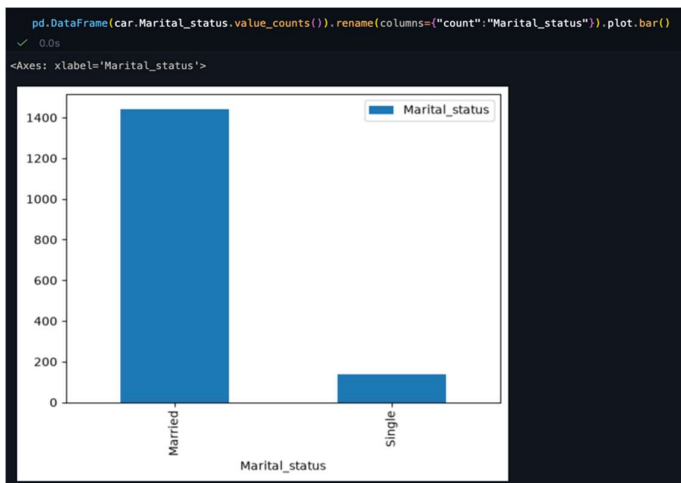
Interpretation

Graduate and Post Graduate customers form the majority of buyers.

Answer

Higher education levels indicate better purchasing capacity and awareness, making these customers attractive targets.

Marital Status



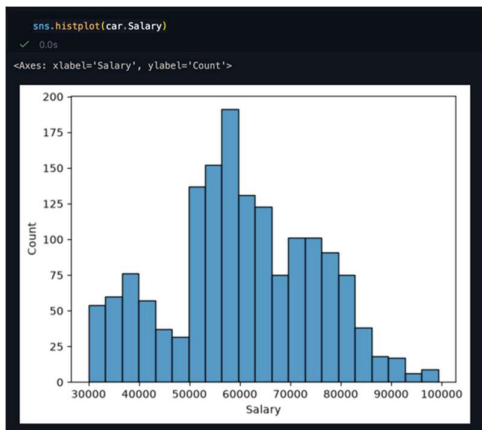
Interpretation

The graph compares married and single customers.

Answer

Married customers represent a larger proportion of vehicle buyers.

Salary Histogram



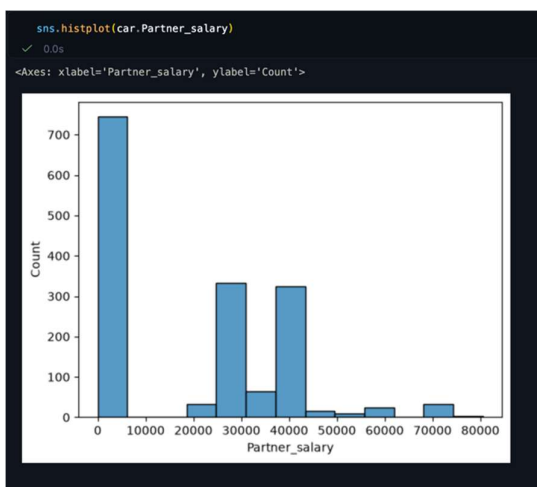
Interpretation

Most customers belong to the middle-income group.

Answer

The company should primarily target middle-income customers while creating premium offerings for high-income groups.

Partner Salary Histogram



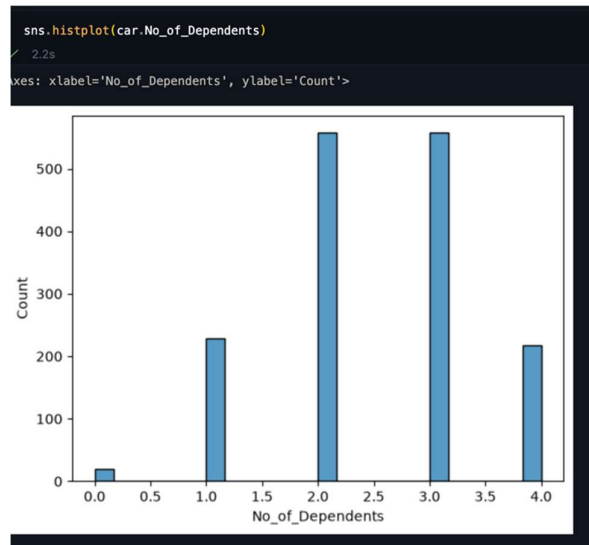
Interpretation

Partner salary contributes significantly to household income.

Answer

Families with higher combined income have greater purchasing power.

Dependents Distribution



Interpretation

Most customers have two or three dependents.

Answer

Family size influences vehicle selection, particularly larger vehicles.

Bivariate Analysis

Gender vs Make

```
car.pivot_table(index="Gender", columns="Make", aggfunc="count")["Age"]
✓ 0.0s
```

Make	Hatchback	SUV	Sedan
Gender			
Female	15	173	141
Male	567	124	561

Objective

To determine whether gender influences vehicle choice.

Interpretation

Male customers purchase more vehicles overall, while female customers show comparatively higher SUV preference.

Answer to Problem Statement

Yes, gender influences vehicle purchasing behavior. Marketing campaigns can be customized for male and female customer segments.

Profession vs Make

```
car.pivot_table(index="Profession", columns="Make", aggfunc="count")["Age"]
```

✓ 0.0s

Make	Hatchback	SUV	Sedan
Profession			
Business	290	89	306
Salaried	292	208	396

Interpretation

Salaried customers mainly purchase Sedans, whereas business customers show interest in Hatchbacks.

Answer

Profession significantly influences vehicle preference.

Education vs Make

```
car.pivot_table(index="Education", columns="Make", aggfunc="count")["Age"]
```

✓ 0.0s

Make	Hatchback	SUV	Sedan
Education			
Graduate	220	117	259
Post Graduate	362	180	443

Interpretation

Post Graduate customers purchase more premium vehicles than Graduates.

Answer

Higher education is associated with increased purchasing capacity.

Marital Status vs Make



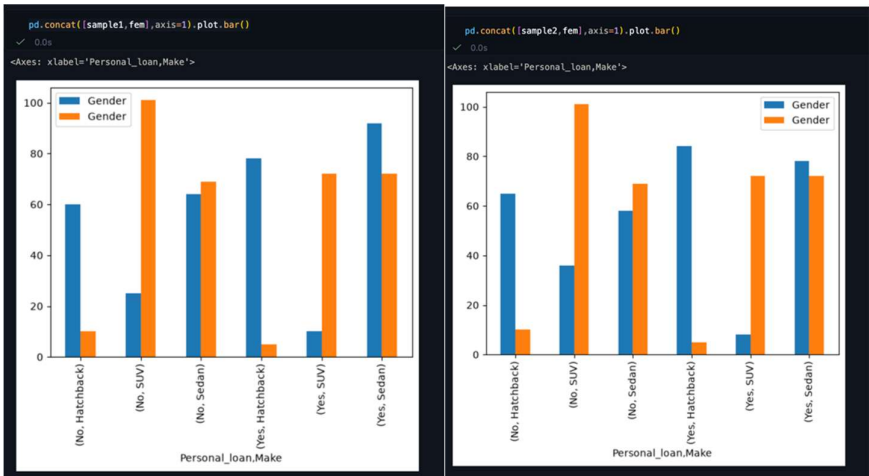
Interpretation

Married customers purchase more vehicles than single customers.

Answer

Family responsibilities influence vehicle buying decisions.

Personal Loan vs Make



Where always orange bar is used for Female and bluebar is used for Male

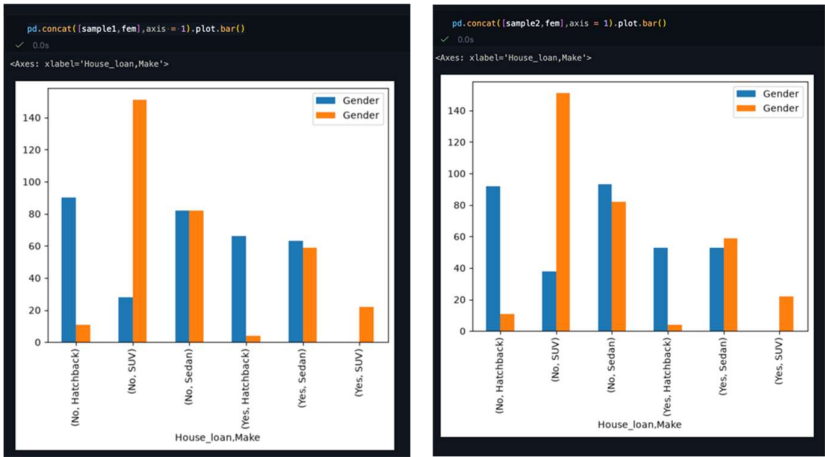
Interpretation

Customers without personal loans purchase more vehicles.

Answer

Existing loan obligations reduce purchasing flexibility.

House Loan vs Make



Interpretation

Customers with house loans tend to purchase fewer premium vehicles.

Answer

Housing expenses impact vehicle purchasing decisions.

Partner Working vs Make



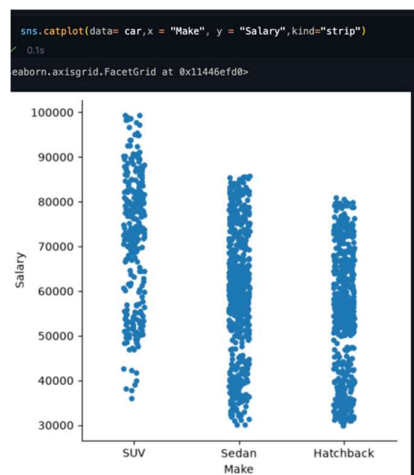
Interpretation

Dual-income families show a higher preference for SUVs and Sedans.

Answer

Families with two earning members have stronger purchasing power.

Salary vs Make



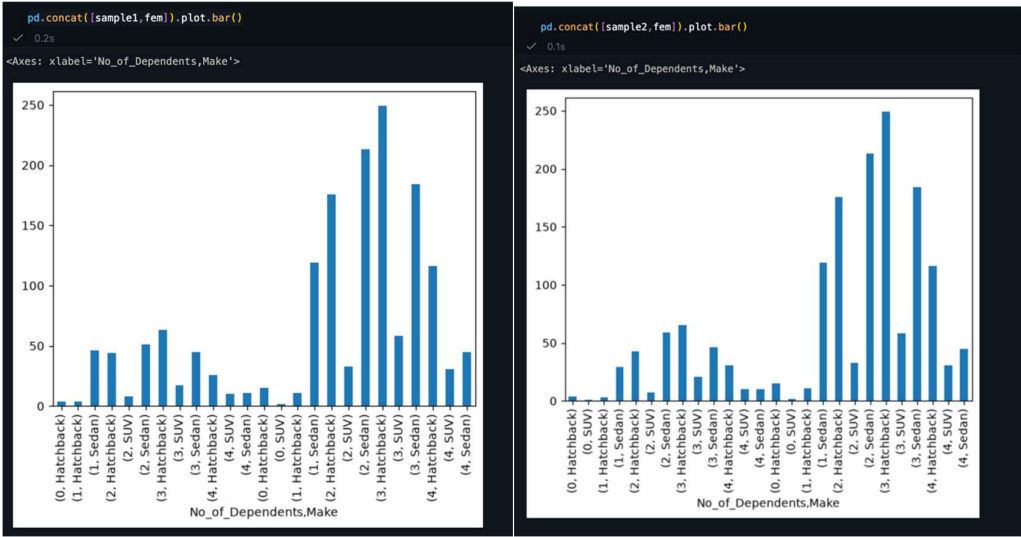
Interpretation

SUV purchases increase with customer salary.

Answer

Income is one of the strongest factors affecting vehicle preference.

Number of Dependents vs Make



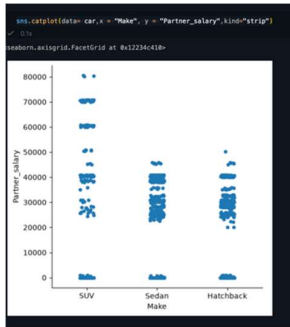
Interpretation

Customers with more dependents prefer family-oriented vehicles.

Answer

Family size should be considered while marketing different vehicle categories.

Partner Salary vs Make



Interpretation

Higher partner salary corresponds with increased purchases of premium vehicles.

Answer

Combined household income is an important predictor of vehicle choice.

8. Answers to the Problem Statement

After performing Exploratory Data Analysis on the Austo Motor Company customer dataset, the following business questions have been answered.

Question 1. Which type of customers purchase Hatchbacks, Sedans, and SUVs?

Answer

The analysis indicates that different customer groups prefer different vehicle categories.

- **Hatchbacks** are mostly preferred by business professionals, customers with lower to middle incomes, and families with more dependents because they are economical and practical.
- **Sedans** are primarily purchased by salaried employees, married customers, and middle-income families due to their balance of comfort and affordability.

- SUVs are generally purchased by customers with higher salaries, postgraduate education, and dual-income households where both partners are working. These customers have greater purchasing power and often prefer premium vehicles.

Business Conclusion:

Vehicle preference changes according to customer income, profession, and family status. Therefore, Austo Motor Company should segment its customers before launching marketing campaigns.

Question 2. Does gender influence vehicle selection?

Answer

Yes. The analysis shows that gender has an impact on vehicle purchasing behavior.

Male customers account for the majority of vehicle purchases across all categories. Female customers also purchase vehicles but show a relatively stronger preference for SUVs compared to Hatchbacks.

Business Conclusion:

The company should create separate promotional campaigns for male and female customers based on their purchasing preferences.

Question 3. Which profession purchases more vehicles?

Answer

Salaried customers purchase more vehicles than business customers.

Among salaried customers, Sedans are the most preferred vehicle category because they offer a balance between comfort and affordability. Business customers show a relatively higher preference for Hatchbacks.

Business Conclusion:

Salaried professionals represent the largest customer segment and should remain the company's primary marketing focus.

Question 4. Does salary affect vehicle preference?

Answer

Yes. Salary has a strong influence on vehicle selection.

Customers with lower salaries mostly purchase Hatchbacks, while middle-income customers generally choose Sedans. Customers with higher salaries are more likely to purchase SUVs because they can afford premium vehicles with additional features.

Business Conclusion:

Income is one of the strongest factors affecting customer purchasing decisions.

Question 5. Do education and marital status influence purchasing behavior?

Answer

Yes.

Customers with postgraduate education purchase more premium vehicles than graduate customers. Married customers purchase more vehicles than single customers because family requirements often influence vehicle ownership.

Business Conclusion:

Education and family status should be considered while designing promotional campaigns.

Question 6. Does having personal or house loans impact buying decisions?

Answer

Yes.

Customers who do not have personal or house loans purchase vehicles more frequently than customers who already have financial liabilities. Existing loans reduce disposable income, making customers less likely to purchase premium vehicles.

Business Conclusion:

Special financing offers and attractive EMI plans can encourage customers with existing loans to purchase vehicles.

Question 7. How does family income affect vehicle choice?

Answer

Family income significantly affects purchasing decisions.

Households with higher combined income (customer salary and partner salary) show greater interest in premium vehicles such as SUVs. Families with lower combined income generally purchase Hatchbacks and Sedans because they are more economical.

Business Conclusion:

Dual-income households are one of the most valuable customer segments for premium vehicle sales.

Question 8. Which customer segment should the company target?

Answer

Based on the analysis, Austo Motor Company should primarily target:

- Salaried professionals
- Middle-income and high-income customers
- Married customers
- Dual-income families
- Postgraduate professionals
- Customers without existing personal or house loans

These customer segments have stronger purchasing power and contribute the largest share of vehicle purchases.

Business Conclusion:

Targeted marketing based on customer segmentation will improve conversion rates and increase overall sales.

9. Business Insights

After analyzing the customer data, the following insights were observed:

- Male customers purchase more vehicles than female customers.
 - Salaried professionals form the largest customer segment.
 - Middle-income customers dominate the customer base.
 - Higher salaries increase SUV purchases.
 - Married customers buy more vehicles than single customers.
 - Customers without loans purchase vehicles more frequently.
 - Dual-income families are more likely to purchase premium vehicles.
 - Education positively influences purchasing behavior.
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10. Recommendations

Based on the analysis, Austo Motor Company should:

1. Focus marketing campaigns on salaried professionals.
 2. Promote SUVs to high-income families.
 3. Offer attractive finance options to middle-income customers.
 4. Introduce family-oriented promotions for married customers.
 5. Create premium packages for dual-income households.
 6. Target female customers with SUV-focused campaigns.
 7. Use customer segmentation for personalized advertising.
 8. Develop loyalty programs for repeat customers.
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11. Conclusion

The customer data analysis successfully identified the major demographic and financial factors influencing vehicle purchases. Salary, profession, marital status, education, and family income play significant roles in determining customer preferences. These findings can help Austo Motor Company make data-driven decisions, improve customer segmentation, and increase sales through targeted marketing strategies.